

Emilie Pereira

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EDUCATION

University of Maryland, College Park

Bachelor of Science in Mechanical Engineering.

Montgomery College

Part-Time General Education Coursework. GPA: 4.0/4.0.

Expected May 2029

Rockville, MD

May 2025

PROJECTS

3D Printed Solar-Lit Home

- Modeled and fabricated a small-scale house in Onshape over one to two months, incorporating 3D printed components and a wood-framed roof; wired a solar panel to power LED lighting throughout the structure.
- Managed the full design-to-fabrication workflow independently, making iterative decisions on component fit, structural assembly, and electrical integration.

Hydraulic Arm

- Engineered and constructed a hydraulic arm for a class-wide competition; outperformed approximately 10 other solo teams to win first place and a \$50 prize.
- Applied fluid mechanics and mechanical linkage principles to design a functional arm capable of competitive performance.

Wooden Bridge Challenge

- Built a 9-inch long, 4-inch high wooden bridge under restricted material constraints; sustained approximately 85 lbs without breaking, outperforming all other entries, the next best of which held 30 lbs.
- Optimized joint placement and load distribution within material limits; the only bridge in the class to remain intact.

Wooden Tower Strength Challenge

- Constructed a wooden tower weighing 0.195 lbs that withstood approximately 90 lbs of applied load; the only tower among approximately 13 competitors that did not break.
- Applied structural design principles to balance material efficiency and load-bearing capacity.

Fire Suppression System

- Developed and programmed a small-scale automated fire detection and suppression system using an Arduino, heat sensor, smoke sensor, and water pump.
- Integrated multiple sensor inputs with programmatic control logic to trigger suppression response automatically upon detection.

Handmade Architectural Home Model

- Designed and constructed a detailed small-scale home model for a class using cardstock, tissue paper, felt, wooden rods, colored paper, cotton, and hot glue; handcrafted the floor plan, interior furniture, and full exterior.
- Applied material selection and assembly techniques to achieve a realistic interior and exterior.

Intro Robotics Builds

- Assembled and programmed a small robot to execute multiple commands including directional movement and line following, using basic control logic and mechanical assembly techniques.

EXPERIENCE

Team Member

June 2025 – August 2026

Playa Bowls

Silver Spring, MD

- Collaborated with team members to prepare and serve food, maintaining quality and customer satisfaction.
- Trained new staff on store procedures and upheld a clean, efficient work environment.

Secretary

November 2023 – June 2024

Pereira Construction

Silver Spring, MD

- Managed daily office operations including scheduling meetings, coordinating calendars, and organizing files.
- Handled incoming calls, emails, and correspondence, routing communications to appropriate departments.
- Maintained organized filing systems for physical and digital records to ensure easy accessibility.

Teacher Assistant Volunteer

September 2021 – June 2022

St. Patrick's Catholic Church

Rockville, MD

- Supervised children to support their health, welfare, and safety in a classroom setting.
- Fostered a consistent, safe, and positive learning environment for students and staff.

SKILLS

Design and Fabrication: CAD (Onshape, Creo Parametric), 3D Printing, Hydraulic Systems, Robotics and Mechanical Assembly, Soldering, Torch Work, Metal Forming

Programming: Embedded and Microcontroller Programming (Arduino)